INCH-POUND

A-A-50104B
October 28, 1992
SUPERSEDING
A-A-50104A
March 12, 1991

COMMERCIAL ITEM DESCRIPTION

RACK, STORAGE, WAREHOUSE

The General Services Administration has authorized the use of this commercial item description.

1. CLASSIFICATION

1.1 Classification. This commercial item description covers storage racks in the following sizes and styles (see 5.2):

Size 1 - 108 inch long beam
Size 2 - 111 inch long beam
Size 3 - 96 inch long beam
Size 4 - 120 inch long beam (for drive through units)

Style A - 36 inch wide upright frame (for wire mesh decking) (rack shelving without pallets)
Style B - 42 inch wide upright frame (for use with pallets)

1.2 CID based part identification number (PIN). A document based PIN to identify sizes and styles of storage racks is included in 5.4. This identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-UXT, Natick, MA 01760-5017.

AMSC N/A FSC 7125

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2. SALIENT CHARACTERISTICS

2.1 Design and construction. Racks shall consist of the following steel components: removable wire mesh decking (for style A racks), upright frame posts, shelf beams with safety locks, front-to-rear shelf supports, welded cross braces, back-to-back connectors, wall spacers, leveling shims, and floor bearing plates. The pallet rack components may be manufactured from hot-rolled steel or cold-formed steel. The design of all hot-rolled steel components shall conform to the American Institute of Steel Construction (AISC), Manual of Steel Construction Allowable Stress Design. Cold-formed steel components shall conform to the American Iron and Steel Institute (AISI), Cold-Formed Steel Construction Manual. The load bearing components shall be designed for easy side-to-side rack assembly or back-to-back connecting during rack erection without the use of special tools. These components shall be constructed to produce a rigid and stable rack capable of withstanding a minimum shelf load of 6,000 pounds. The upright frame unit for size 1 and size 3 racks shall have a minimum load capacity of 24,000 pounds and size 2 and size 4 racks shall have a minimum load capacity of 31,000 pounds. An assembled rack unit shall consist of two upright frames attached to three shelf units. All steel components shall be galvanized in accordance with ASTM A 123.

2.2 Upright frame. The upright frame shall be welded construction and shall consist of two vertical posts, minimum 3 inch by 3 inch box channel or tabular column. For tension members, the posts shall be braced by a minimum of five braced panels consisting of "X" type bars. For compression members, the posts shall be braced by a minimum of five braced panels consisting of diagonal angles or channels or braced by five horizontal and four X-brace panel construction. The braces shall be located between two horizontal bars for each brace and the posts shall be secured to the floor bearing plates. Frames shall be interchangeable in any assembled position. Each post shall have openings to accommodate shelf-beam connectors at corresponding heights. The upright frames shall be from 204 to 207 inches high, and shall all be of equal length. Intermediate upright frame units shall be approximately 180 inches high with a minimum of two bracing panels (see 5.2).

2.3 Shelf unit. The shelf unit shall consist of two 108 inch long load beams for the size 1 racks, two 111 inch long beams for the size 2 racks, two 96 inch long beams for the size 3 racks, and two 120 inch long beams for the size 4 racks. The ends of the beams shall have a multiple prong slip fit connector to easily fasten to post openings. Each beam-to-post connector shall have a safety lock which will lock the beam to the posts. The beams shall be 4 to 5 inches high by 2-1/2 to 3 inches deep.

2.3.1 Front to rear shelf supports. Each shelf unit of the style B rack shall be provided with four front-to-rear supports, a minimum of 3 inches wide by 1-5/8 to 2 inches deep. Each front-to-rear support bar shall fit within a 42 inch upright frame and shall withstand a minimum safe load of 1500 pounds. Front to rear supports shall be prevented from lateral movement

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by means of a formed tab or mechanical connection that locks into slots punched in the shelf beams, or by overlap flanges that are mechanically fastened to the beams.

2.3.2 Wire Mesh Decking. Two sections of wire mesh decking of equal length shall be provided for each shelf unit of the style A racks. The wire mesh decking shall be constructed of 2 gauge minimum steel wire, and shall have a 2 inch by 4 inch or a 2 inch by 2 inch diamond or square shaped meshwork as specified (see 5.2). For pallet racks with box type load beams, two minimum 14 gauge flared channel supports a minimum of 2 inches wide by 1-3/8 inches deep shall be welded to each section of wire mesh decking. For pallet racks with step type load beams, two minimum 12 gauge "U" shaped supports a minimum of 1 inch wide by 1-3/8 inches deep shall be welded to each section of wire mesh decking. The front and rear edges of the decking shall be waterfall design (flanged) to fit over load beams. The wire mesh decking shall be removable from the rack shelf. The minimum shelf capacity of each section of deck shall be 3000 pounds.

2.4 Back-to-back connectors. A sufficient amount of back-to-back connectors shall be furnished to securely attach the backs of the units to form a double row of racks. The connectors provided shall be either the slip-on boltless type or the bolt, washer, and nut variety as specified (see 5.2). The connectors shall measure approximately 12 inches long and shall have strength to withstand all forces applied to the connected rack unit during normal loading and unloading operations.

2,5 Leveling shims and wall spacers. A total of 1/8 inch per column of leveling shims shall be provided with each rack unit. The shim shall be a minimum of 12 gauge steel and be the same size as the floor bearing plate. The shim shall anchor or lock with the floor bearing plate in such a manner as not to separate during normal storage operations. A sufficient amount of wall spacers shall be provided for the purpose of connecting rack to wall for added stability while maintaining desired spacing.

2.5.1 Anchor bolts. As a minimum, one drilled in place wedge type anchor bolt shall be supplied with each side column. Bolts shall be a minimum of 1/2 inch diameter with an embedment into the concrete floor of at least 2-1/2 inches.

2.6 Corner protectors. Each exposed corner of a row of racks shall be furnished with a protective corner bumper (see 5.2). The steel bumper shall rise a minimum of 12 inches from the floor and measure a minimum of 3/16 inch thick by 3 inches by 3 inches. Corner bumpers shall not be welded to the upright. The bumpers shall be painted yellow, conforming to Color No. 13538 of FED-STD-595, and shall be lagged to the floor. Corner bumpers shall have a minimum of two anchor bolts.

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3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3.3 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

4. PACKAGING

4.1 Packaging. The storage rack shall be packaged, packed,and marked in accordance with the domestic or overseas requirements of ASTM D 3951, as applicable.

5. NOTES

5.1 Intended use. The storage racks are for indoor use. Style A racks are 36 inches wide and have wire mesh decking for non-palletized storage. Style B racks are 42 inches wide and are used only for palletized storage.

5.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Size and style of rack required (see 1.1).
c. Number of upright frames required (see 2.2).
d. Type of shaped meshwork required (see 2.3.2).
e. Type of back-to-back connectors required (see 2.4).
f. Number of corner protectors required (see 2.6).

5.3 Sources of documents.

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5.3.1 Sources for non-Government association documents.

ASTM A 123 - Zinc (Hot Galvanized) Coatings on Products Fabricated from Rolled, Pressed, and Forged Steel Shapes, Plates, Bars, and Strip

ASTM D 3951 - Standard Practice for Commercial Packaging, are available from.

The American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103-1187

AISC - Manual for Steel Construction, Allowable Stress Design, is available from:

The American Institute of Steel Construction (AISC)
1 East Wacker Drive
Suite 3100
Chicago, IL 60601-2001

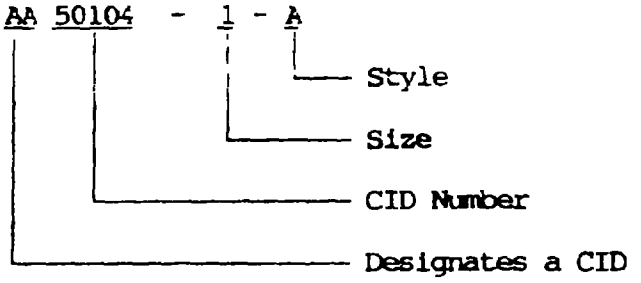
AISI - Specification for the Design of Cold-Formed Steel Structural Members, is available from:

The American Iron and Steel Institute (AISI)
1133 15th N.W. Street
Suite 300
Washington, DC 20005-2701

5.3.2 Sources of Government documents. Copies of military and federal documents are available from:

Standardization Documents Order Desk
Building 4D
700 Robins Avenue
Philadelphia, PA 19111-5094

5.4 Part identification number (PIN). The PINs to be used for the storage racks required by this CID are created as follows:



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MILITARY INTERESTS:

Custodians

Army - GL
Air Force - 99

Review Activity

Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY

Army - GL

Project 7125-0139

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